STROKE DIAGNOSE PREDICTION

Introduction

Introduction: Modeling Uncertainty in Healthcare

- The Problem: Stroke is a critical medical event, but its occurrence is stochastic (random) rather than deterministic.
- The Goal: To quantify the risk. We do not just want a "Yes/No" prediction; we need to estimate the Conditional Probability of a stroke given a patient's health metrics:
- P(Stroke | Age,BMI,Heart disease,...)

Methodology: Entropy

Quantifying Uncertainty with Entropy
The Goal: We want to split "messy" data (impurity) into "pure" nodes
Measures the "surprise" of a class ci.

$P(c_i)$ is the probability of class Ci

$$I(c_i) = \log_2\left(\frac{1}{P(c_i)}\right) = -\log_2 P(c_i)$$

Entropy is the Expected Value of Information Content.

$$\mathbb{E}[X] = \sum x \cdot P(x)$$

$$\mathbb{E}[I(Y)] = \sum_{i=1}^{k} P(Y = c_i) \cdot \underbrace{I(Y = c_i)}_{\text{Giá trị}}$$

Entropy Formula

$$\boxed{H(Y) = \sum_{i=1}^{k} P(c_i) \cdot (-\log_2 P(c_i))}$$

Algorithm: Finding the Optimal Split Point (Decision Tree)

Step 1: Consider all possible split points
Call F={X1,X2,...,Xm} is a set of random variables (input characteristics) that can be used for observation.

Step 2: Calculate Entropy of each branch (Conditional Entropy)
After assuming dividing the data according to the characteristic X

$$H(Y|X) = \sum_{v \in Values(X)} \frac{|D_v|}{|D|} H(D_v)$$

Where: $\frac{|D_v|}{|D|}$ represents the probability of falling into branch v, and $H(D_v)$ is the Entropy at that branch.

Step 3: Calculate Information Gain (Delta Expected Value)
We compare the initial impurity with the impurity after splitting to determine the amount of "information" gained.

$$IG(Y, X) = H(Y) - H(Y|X)$$

$$IG(Y, X) = \underbrace{\mathbb{E}[-\log P(Y)]}_{\text{Initial Entropy } H(Y)} - \underbrace{\mathbb{E}_X[\mathbb{E}_{Y|X}[-\log P(Y|X)]]}_{\text{Entropy after knowing } X}$$

Step 4: Select the point with the highest Information Gain

$$X^* = \underset{X \in \mathcal{F}}{\operatorname{argmax}} \left(H(Y) - \sum_{v \in Values(X)} \frac{|D_v|}{|D|} H(Y|X = v) \right)$$

Step 5: Recursive Partitioning
We treat each resulting branch as a new, independent dataset.
We repeat Steps 1-4 for each of these branches

How to find the conditional probability of stroke, given a patient's health metrics x.

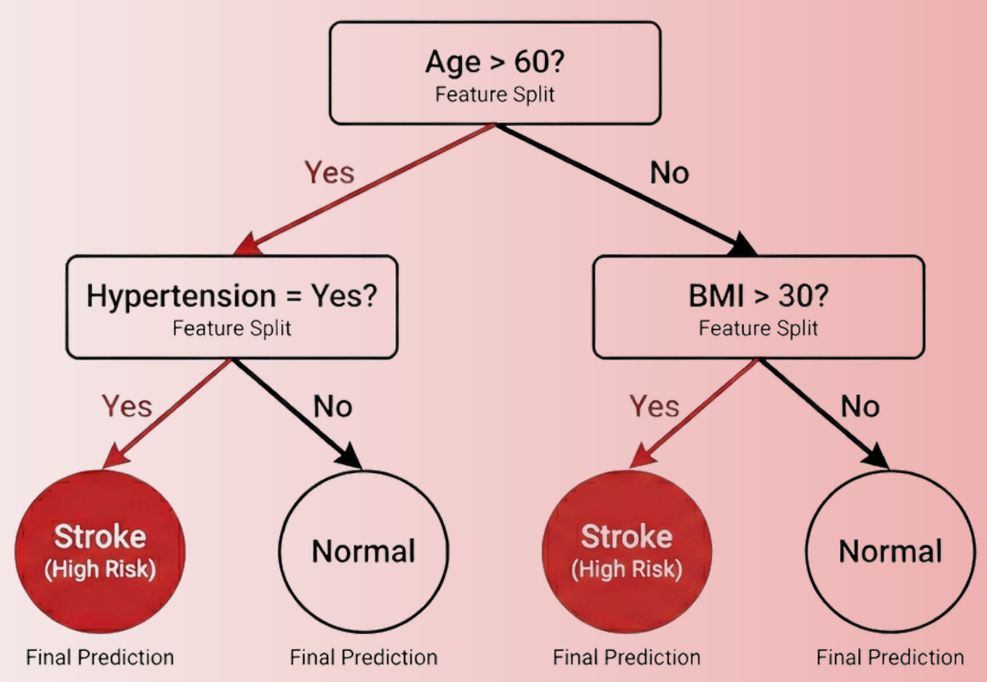
Concept: Instead of relying on a single Decision Tree, we construct a Random Forest consisting of N independent trees (e.g., N=100)

Definition: We define each tree in the forest as a **Random Variable**, denoted as T_i.

- Let $T_i(x) = 1$ if the i-th tree predicts "Stroke".
- Let $T_i(x) = 0$ if the i-th tree predicts "No Stroke".

Expected Value (the mean) of our random variables Ti.

$$\hat{P}(Y = 1|x) = \frac{1}{N} \sum_{i=1}^{N} T_i(x)$$

Eg. If 85 out of 100 trees vote "Stroke", P^=0.85 (or 85%)

Example of a simple decision tree